MIT 18.06SC - 3.1

Problem 25.1

Step 1:-

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} i \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ i \end{bmatrix} = i\begin{bmatrix} i \\ 1 \end{bmatrix}$$

Step 2:-

$$\begin{bmatrix} i & 1 \end{bmatrix}\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} i \\ 1 \end{bmatrix} = 0 = i\begin{bmatrix} i & 1 \end{bmatrix}\begin{bmatrix} i \\ 1 \end{bmatrix}$$

Step 3:-

$$\lambda = \frac{\begin{bmatrix} i & 1 \end{bmatrix}\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} i \\ 1 \end{bmatrix}}{\begin{bmatrix} i & 1 \end{bmatrix}\begin{bmatrix} i \\ 1 \end{bmatrix}}$$

From step 2 and step 3 it follows that $\lambda = 0/0$ which is undefined. The proof is assuming that $x^T x \neq 0$. i.e. it is assuming that x has no complex entries.

Problem 25.2

a) No. Let $A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 2 \\ 2 & 4 \end{bmatrix}$.

Then $AB = \begin{bmatrix} 17 & 10 \\ 9 & 10 \end{bmatrix}$ which is not symmetric.

Yes, they form a group.

b) $BB^T = I$ so $B^T = B^{-1}$ and since B is square, $B^TB = I$ so B^T (i.e. B^{-1}) is orthogonal.

$B^TB = I$ and $A^TA = I$ if A and B are any multiplyable matrices. $(AB)^T(AB) = A^TB^TAB$

$= B^TIB = B^TB = I$

Hence AB is orthogonal.

c) Yes they form a group. Let e^{xA} and e^{yA} be any exponentials of a fixed matrix A. Then $e^{xA} \cdot e^{yA} = e^{xA+yA} = e^{A(x+y)}$ which is also of the form e^{tA}. Also, $(e^{xA})^{-1} = e^{-xA}$ which is also of the form e^{tA}.

d) Yes, they form a group. Let A and B be any two matrices such that $\det(A) = \det(B) = 1$. Then $\det(AB) = \det(A) \times \det(B) = 1 \times 1 = 1$ and $\det(A^{-1}) = 1/\det(A) = 1/1 = 1$.